

HAZOP ANALYSIS FOR REFRIGERATION SYSTEM

ME-325
TEAM- REFLOSAFE

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HAZOP Methodology

1.HAZOP Methodology

1.1 Study Approach

The HAZOP study employs the systematic guide-word technique as defined in IEC 61882 and follows IIAR guidelines. The analysis examines deviations from design intent using established guide words applied to key process parameters. *References* [8][7]

Standard Guide Words:

- **NO/NOT:** Complete absence of design intent
- **MORE:** Quantitative increase beyond normal parameters
- **LESS:** Quantitative decrease below normal parameters
- **AS WELL AS:** Additional unintended materials or conditions
- **PART OF:** Partial achievement of design intent
- **REVERSE:** Opposite of intended direction or effect
- **OTHER THAN:** Complete substitution of intended conditions

References [8][7]

Process Parameters Analyzed:

- **Flow:** Liquid and vapor flow rates
- **Pressure:** Operating pressures throughout system
- **Temperature:** Process temperatures and thermal conditions
- **Level:** Liquid levels in vessels and separators
- **Composition:** Refrigerant purity and contamination
- **Time:** Sequence and duration of operations

References [8][7]

1.2 Risk Assessment Framework

The risk assessment utilizes a 5×5 matrix combining severity and likelihood ratings to generate risk scores from 1-25. Risk categories are defined as follows:

- **Low Risk (1-4):** Acceptable with routine monitoring
- **Medium Risk (5-9):** Requires management attention and controls
- **High Risk (10-16):** Immediate action required
- **Critical Risk (17-25):** Unacceptable, requires immediate shutdown

References [10][9]



Figure 1:HAZOP risk matrix

HAZOP Risk Assessment Matrix showing the relationship between severity and likelihood for ammonia refrigeration system hazards.

HAZOP Analysis for Ammonia Refrigeration System

This enhanced HAZOP study provides a detailed safety analysis of the ammonia refrigeration system, incorporating industry best practices, regulatory requirements, and comprehensive risk assessment methodologies. The analysis follows IAR standards and international HAZOP guidelines to ensure thorough hazard identification and operability assessment.

2. System Description and Scope

2.1 Process Overview

The ammonia refrigeration system operates as a closed-loop mechanical compression system utilizing anhydrous ammonia (NH₃, R-717) as the primary refrigerant. The system maintains optimal temperatures for industrial cooling applications while adhering to strict safety protocols due to ammonia's toxic and corrosive properties. *Reference [34][1]*

Key System Parameters:

- **Operating Pressure Range:** 93.7 kPa to 1,351 kPa
- **Temperature Range:** -35°C to +35°C
- **Ammonia Charge:** Approximately 3,500 kg

- **System Type:** Two-stage compression with liquid recirculation
- **Refrigerant Grade:** 99.98% purity anhydrous ammonia

References[2][3]

2.2 Major System Components

Primary Equipment:

- **Compressor Units:** Reciprocating and screw compressors with VFD control
- **Evaporators:** Shell-and-tube heat exchangers with flooded design
- **Condensers:** Air-cooled and water-cooled units
- **Receivers:** High and low-pressure storage vessels
- **Piping Network:** Carbon steel piping with appropriate wall thickness
- **Control Systems:** Automated process control with safety interlocks

3. Hazard Analysis by Node

3.1 Node 01 - Ammonia Compressor

System Function: Compress ammonia vapor from evaporator suction pressure to condenser discharge pressure while maintaining proper oil separation and thermal management. *References* [4][5]

Table 1:Hazard analysis for compressor

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
NO	Flow	No suction flow	Upstream valve closure, line blockage, pump failure	Compressor damage, system shutdown, loss of refrigeration	Suction pressure alarms, low-pressure cutout, anti-surge protection	4	3	12	Install redundant suction pressure monitoring, upgrade anti-surge system
MORE	Pressure	High discharge pressure	Condenser fouling, high ambient temperature, blocked discharge	Compressor overload, safety valve lifting, potential rupture	High-pressure cutout, relief valves, pressure monitoring	5	2	10	Implement condenser cleaning schedule, upgrade relief capacity
LESS	Temperature	Low suction temperature	Expansion valve malfunction, low heat load	Liquid slugging, compressor damage, bearing failure	Suction superheat monitoring, liquid injection protection	4	3	12	Install liquid detection system, upgrade superheat control

AS WELL AS	Contamination	Oil carryover	Oil separator failure, high oil level	Reduced heat transfer, system contamination, valve damage	Oil level monitoring, oil separator maintenance	3	3	9	Upgrade oil separation system, implement oil analysis program
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3.2 Node 02 - Storage Tank/Receiver

System Function: Store liquid ammonia inventory to accommodate system variations and provide surge capacity while maintaining proper pressure and level control. *References* [4][6]

Table 02: Hazard analysis for storage tank/receiver

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
MORE	Level	High liquid level	Control valve failure, level instrument malfunction	Liquid carryover, reduced surge capacity, potential overflow	Level alarms, high-level cutout, visual indicators	3	2	6	Install redundant level measurement, upgrade dump systems
MORE	Pressure	Overpressure	Thermal expansion, blocked outlet, relief valve failure	Vessel rupture, catastrophic ammonia release	Relief valves, pressure monitoring, thermal protection	5	2	10	Upgrade relief system, install thermal insulation
NO	Ventilation	No pressure equalization	Blocked vent, valve closure	Pressure differential, vessel deformation	Pressure equalization lines, vacuum protection	3	2	6	Install vacuum breakers, upgrade vent systems

3.3 Node 03 - Piping System

System Function: Transport ammonia between system components while maintaining pressure integrity and preventing external leakage. *References* [1][11]

Table 03: Hazard analysis for piping system

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
NO	Flow	Complete flow blockage	Ice formation, foreign material, valve malfunction	Pressure buildup, system imbalance, component failure	Flow indicators, pressure monitoring, bypass systems	4	2	8	Install heat tracing, upgrade filtration systems

AS WELL AS	Leakage	External ammonia release	Pipe corrosion, joint failure, thermal stress	Toxic exposure, environmental release, evacuation	Leak detection, emergency shutdown, ventilation	4	3	12	Upgrade leak detection sensitivity, install emergency isolation
MORE	Pressure	Hydraulic shock	Rapid valve closure, pump start/stop, thermal effects	Pipe rupture, fitting failure, system damage	Pressure relief, shock arresters, slow-acting valves	4	2	8	Install shock suppressors, upgrade pipe supports
OTHER THAN	Material	Wrong pipe material	Installation error, maintenance mistake	Corrosion, failure, contamination	Material certification, inspection procedures	5	1	5	Implement material traceability, color coding system

3.4 Node 04 - Valves (General)

System Function: Control and isolate ammonia flow throughout the system with reliable operation under all conditions. *References* [12][13]

Table 04: Hazard analysis for valves

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
NO	Operation	Fail to open	Actuator failure, power loss, mechanical binding	Process disruption, pressure buildup, emergency situation	Manual overrides, position feedback, backup power	4	2	8	Install redundant actuators, upgrade position monitoring
PART OF	Sealing	Internal leakage	Seat damage, foreign material, thermal cycling	Cross-contamination, efficiency loss, pressure loss	Leak testing, pressure monitoring, maintenance schedule	3	3	9	Upgrade seat materials, implement predictive maintenance
AS WELL AS	Leakage	External stem leakage	Packing failure, thermal expansion, over-tightening	Ammonia release, toxic exposure, environmental impact	Leak detection, fugitive emission monitoring, PPE	4	3	12	Upgrade packing systems, install leak collection
REVERSE	Flow	Backflow	Check valve failure, pressure reversal	Contamination, equipment damage, process upset	Check valves, flow indicators, pressure monitoring	3	2	6	Install redundant check valves, upgrade flow monitoring

3.5 Node 05 - Safety Relief Valves

System Function: Provide overpressure protection for system components and prevent catastrophic failures while managing ammonia discharge safely. *References* [1][14][15]

Table 05: Hazard analysis for safety relief valves

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
NO	Operation	Fail to open	Valve sticking, inlet blockage, spring failure	Overpressure, vessel rupture, catastrophic release	Redundant relief valves, pressure monitoring, testing schedule	5	2	10	Install dual relief systems, upgrade testing frequency
REVERSE	Sealing	Fail to reseal	Seat damage, foreign material, spring fatigue	Continuous venting, toxic cloud, pressure loss	Downstream monitoring, isolation valves, maintenance	4	3	12	Upgrade seat materials, install downstream isolation
LESS	Capacity	Insufficient relief	Undersized valve, blocked discharge, multiple scenarios	Overpressure, system damage, safety compromise	Capacity calculations, regular testing, multiple valves	5	2	10	Recalculate relief requirements, upgrade discharge systems
MORE	Pressure	Premature opening	Low set pressure, thermal effects, vibration	Nuisance trips, ammonia loss, operational disruption	Pressure monitoring, vibration damping, thermal protection	2	3	6	Install vibration dampers, upgrade thermal protection

3.6 Node 06 - Leak Detection System

System Function: Provide early warning of ammonia releases to protect personnel and enable rapid response to minimize consequences. *References* [16][18]

Table 06: Hazard analysis for leak detection system

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
NO	Detection	System failure	Sensor failure, power loss, calibration drift	Undetected leaks, delayed response, increased exposure	Redundant sensors, backup power, alarm systems	4	3	12	Install triple redundancy, upgrade power backup

MORE	Sensitivity	False alarms	Sensor drift, environmental interference, contamination	Nuisance shutdowns, reduced confidence, maintenance burden	Alarm management, sensor maintenance, filtering	2	4	8	Implement smart sensors, upgrade alarm management
LESS	Coverage	Inadequate monitoring	Sensor placement, blind spots, range limitations	Missed leaks, exposure risk, delayed evacuation	Multiple sensor types, coverage analysis, portable detectors	4	2	8	Conduct coverage study, install additional sensors
OTHER THAN	Response	Wrong action	Logic errors, communication failure, training gaps	Inappropriate response, delayed evacuation, safety compromise	Training programs, response procedures, communication systems	4	2	8	Upgrade control logic, enhance training programs

3.7 Node 07 - Ventilation System

System Function: Maintain safe atmosphere in equipment areas and provide emergency ventilation to prevent dangerous accumulations of ammonia vapor. *References* [18][19]

Table 07: Hazard analysis for ventilation system

Guide Word	Parameter	Deviation	Potential Causes	Consequences	Existing Safeguards	S	L	R	Recommendations
NO	Ventilation	Complete failure	Fan failure, power loss, duct blockage	Ammonia accumulation, toxic atmosphere, evacuation required	Backup fans, emergency power, monitoring systems	4	3	12	Install redundant fans, upgrade emergency power
LESS	Airflow	Reduced capacity	Partial blockage, damper malfunction, motor degradation	Poor dilution, elevated concentrations, exposure risk	Airflow monitoring, pressure differential, maintenance	3	3	9	Install flow monitoring, upgrade maintenance program
REVERSE	Direction	Wrong airflow	Damper failure, wind effects, pressure imbalance	Contamination spread, reduced effectiveness, exposure risk	Directional monitoring, pressure control, damper maintenance	3	2	6	Install directional sensors, upgrade damper systems
AS WELL AS	Contamination	System contamination	Duct corrosion, filter failure, external sources	Reduced effectiveness, equipment damage, maintenance issues	Filter systems, corrosion protection, monitoring	2	3	6	Upgrade filtration, implement corrosion control

4. Safety Standards and Compliance

4.1 Regulatory Framework

IIAR Standards Compliance:

- **ANSI/IIAR 2-2021:** Design of Safe Closed-Circuit Ammonia Refrigeration Systems
- **ANSI/IIAR 6-2019:** Inspection, Testing, and Maintenance Requirements
- **ANSI/IIAR 9-2020:** Minimum System Safety Requirements for Existing Systems

References [20][21][12]

OSHA Requirements:

- **29 CFR 1910.119:** Process Safety Management
- **29 CFR 1910.1000:** Permissible Exposure Limits (50 ppm TWA)
- **29 CFR 1910.38:** Emergency Action Plans

References [22][23]

4.2 Exposure Limits and Health Effects

Critical Exposure Thresholds:

- **Odor Threshold:** 5-20 ppm (detectable)
- **OSHA PEL:** 50 ppm (8-hour TWA)
- **ACGIH TLV:** 25 ppm (8-hour TWA)
- **STEL:** 35 ppm (15-minute exposure)
- **IDLH:** 300 ppm (immediately dangerous to life and health)

References [22][24][25]

Health Effects by Concentration:

- **25-50 ppm:** Mild irritation, detectable odor
- **100-300 ppm:** Severe irritation, respiratory distress
- **300-500 ppm:** Immediate evacuation required
- **>500 ppm:** Life-threatening exposure, potential fatality

References [22][24][26]

5. Emergency Response Procedures

5.1 Leak Response Protocol

Immediate Actions:

1. **Detection and Alarm:** Automatic detection triggers building evacuation alarm
2. **Evacuation:** Personnel move upwind and uphill from leak source
3. **Isolation:** Emergency shutdown systems activated remotely
4. **Notification:** Emergency services contacted
5. **Assessment:** Trained personnel assess leak magnitude and wind direction

References [19][27][28]

Personal Protective Equipment:

- **SCBA:** Required for entry into affected areas
- **Chemical Suits:** Level B protection for leak investigation
- **Emergency Eyewash:** Available within 25 feet of work areas
- **Escape Respirators:** Positioned at strategic locations

References [28][29]

5.2 Evacuation Procedures

Evacuation Zones:

- **Immediate Danger Zone**
- **Precautionary Zone**
- **Shelter-in-Place**
- **Assembly Points:** Designated locations upwind of facility

References [19][27][30]

6. Maintenance and Inspection Requirements

6.1 Preventive Maintenance Schedule:

- Daily Inspections
- Weekly Inspections
- Monthly Inspections
- Annual Inspections

7. Training and Competency Requirements

7.1 Personnel Qualifications

- Operators
- Maintenance Personnel

References [29][32]

7.2 Emergency Response Training

- Monthly Drills
- Annual Training

8. Recommendations

This comprehensive HAZOP analysis has identified significant safety improvements needed across all system nodes. The analysis reveals that while the current system incorporates basic safety measures, enhanced protection systems are required to meet current industry standards and regulatory requirements.

Critical Improvements Required:

1. **Enhanced leak detection systems** with redundant sensors and improved coverage
2. **Upgraded emergency ventilation** with backup power and continuous monitoring
3. **Improved relief system design** with dual protection and enhanced capacity
4. **Advanced process monitoring** with predictive maintenance capabilities

9. References

This analysis incorporates requirements from current industry standards including ANSI/IIAR 2-2021, OSHA Process Safety Management regulations, and international best practices for ammonia refrigeration system safety. The recommendations align with proven industry practices and regulatory expectations for comprehensive process hazard analysis.

Key Standards Referenced:

- ANSI/IIAR 2-2021: Design of Safe Closed-Circuit Ammonia Refrigeration Systems
- ANSI/IIAR 6-2019: Inspection, Testing, and Maintenance Requirements
- ANSI/IIAR 9-2020: Minimum System Safety Requirements
- OSHA 29 CFR 1910.119: Process Safety Management
- IEC 61882: Hazard and Operability Studies (HAZOP Studies)

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